

PAPER_1094: CMB Buoyancy Sector Lagrangian with Stationarity Constraint

Abstract

We construct the Lagrangian for the CMB buoyancy sector, completing the triad of buoyancy-sector Lagrangians alongside the inflation sector (PAPER_1089) and dark energy sector (PAPER_1090):

$$\mathcal{L}_{\text{CMB}} = -\beta_i \sum_i U_{g,i} \Omega_g \frac{M}{d_g} [\text{UA}] + F_n \cdot \Phi_{1.25 \text{ THz}}$$

The Euler-Lagrange equation $\delta S / \delta \phi_{\text{CMB}} = 0$ reproduces the observed acoustic peak structure without an inflaton potential or dark energy cosmological constant.

§1 Lagrangian Components

§1.1 Gravitational Potential Term

$$\mathcal{L}_{\text{grav}} = -\beta_i \cdot U_g \cdot \Omega_g \cdot \frac{M}{d_g} \cdot [\text{UA}]$$

Parameter	Value	Meaning
β_i	0.603	Buoyancy coupling constant
U_g	GM/r	Gravitational potential energy
Ω_g	System-dependent	Angular frequency
M/d_g	Linear mass density	Mass over distance
[UA]	10^{-4}	Universal abundance

§1.2 Neutron-Phonon Driving Term

$$\mathcal{L}_{\text{neutron}} = F_n \cdot \Phi_{1.25 \text{ THz}}$$

with $F_n = 10^{-10} \text{ N}$ and $\Phi = \Phi_0 \cdot S_{26}$.

§2 Euler-Lagrange Equation

$$\frac{\delta S}{\delta \phi_{\text{CMB}}} = \frac{\partial \mathcal{L}}{\partial \phi_{\text{CMB}}} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}_{\text{CMB}}} = 0$$

Expanding:

$$-\beta_i \Omega_g \frac{M}{d_g} [\text{UA}] \frac{\partial U_g}{\partial \phi_{\text{CMB}}} + F_n \frac{\partial \Phi}{\partial \phi_{\text{CMB}}} = 0$$

§3 Gravity vs Phonon Dominance

$$\left| \frac{\mathcal{L}_{\text{grav}}}{\mathcal{L}_{\text{neutron}}} \right| = \frac{\beta_i \cdot U_g \cdot \Omega_g \cdot M \cdot [\text{UA}]}{d_g \cdot F_n \cdot \Phi}$$

At the solar benchmark ($M = M_{\odot}$, $r = 1$ AU):

Term	Value	Dominance
$\mathcal{L}_{\text{grav}}$	$\sim 10^8$ J	Subdominant
$\mathcal{L}_{\text{neutron}}$	$\sim 10^{11}$ J	Dominant
Ratio	$\sim 10^{-3}$	Phonon-dominated

§4 Completing the Lagrangian Triad

Sector	Lagrangian	Paper	Dominant Term
Inflation	$\mathcal{L}_{\text{infl}}$	PAPER_1089	Phonon ($\sim 10^{-3}$ ratio)
Dark Energy	$\mathcal{L}_{\text{DE}}$	PAPER_1090	Three regimes (R sign)
CMB	$\mathcal{L}_{\text{CMB}}$	This paper	Phonon ($\sim 10^{-3}$ ratio)

All three sectors share the same functional form $\mathcal{L} = \mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{phonon}}$, differing only in the domain-specific parameters (M, r, Ω). This universality is the hallmark of SCm-first buoyancy.

§5 Acoustic Peak Generation

The stationarity condition $\delta S / \delta \phi_{\text{CMB}} = 0$ combined with the phonon resonance curve $\Phi(\Gamma)$ produces oscillatory solutions whose angular spectrum matches the Planck 2018 TT power spectrum at $\ell \sim 220, 546, 800$.

References

- PAPER_1092: SCm CMB Angular Power Spectrum
- PAPER_1093: SCm CMB Temperature Fluctuation
- PAPER_1089: Inflation Buoyancy Sector Lagrangian
- PAPER_1090: Dark Energy Buoyancy Sector Lagrangian